Bole (James) Pan

269 Vassar Street, Cambridge, MA 02139
+1 (646)-812-0398 | bolepan@mit.edu | LinkedIn | GitHub | Google Scholar | Website

EDUCATION

Massachusetts Institute of Technology, Harvard-MIT Health Sciences and Technology — Expected May 2030
Doctor of Philosophy, Medical Engineering and Medical Physics; GPA: 5.00/5.00

Columbia University in the City of New York, Columbia College — Sept. 2020 – May 2024
Bachelor of Arts, Computer Science, *Summa cum Laude*, Phi Beta Kappa; GPA: 4.08/4.00

Relevant Coursework: Machine Learning, Deep Learning, Real Analysis, Probability Theory, Statistical Inference, Stochastic Systems, Ordinary Differential Equations, Numerical Methods, Dynamical Systems, Stimulations and Modeling, Linear Algebra, Geometric Data Analysis, Analysis of Algorithms, Systems Programming in C, Intensive Organic Chemistry, Human Pathology, Cellular and Molecular Immunology, Genetics, Cardiovascular Pathophysiology, Respiratory Pathophysiology

RESEARCH EXPERIENCE

MIT, Institute of Medical Engineering and Science | Ragon Institute | Broad Institute
Graduate Student Researcher; Principal Investigator: Prof. Alex Shalek — Apr. 2025 – Present

- Investigating how pre-cancerous hepatocytes evolve within the injured liver environments, using spatial transcriptomics, computational methods, and experimental models to inform early liver cancer risk stratification and prevention.

Broad Institute of MIT and Harvard, Cancer Genome Computational Analysis Group
Rotating Graduate Student; Advisor: Prof. Gad Getz — Nov. 2024 – Feb. 2024

- Developed a method integrating AlphaFold-predicted protein structures with somatic mutation data to identify novel cancer driver genes through 3D mutation clustering, with top hits validated experimentally.

Max Planck Institute for Biology of Ageing
Undergraduate Researcher; Advisor: Prof. Ron Jachimowicz — May 2024 – Aug. 2024

- Developed analysis pipelines for whole exome and transcriptomic sequencing data of relapsed/refractory mantle cell lymphoma patients, with the aim of identifying features related to treatment responses.

Memorial Sloan Kettering Cancer Center, Computational and Systems Biology Program
Undergraduate Researcher; Advisor: Prof. Dana Pe’er — Sept. 2023 – May 2024

- Modeling chromatin structural changes with single-cell ATAC-seq data. Employed geometric data analysis and statistical inference to reveal nucleosome kinetics during prostate cancer neuroendocrine transformation.

INDUSTRY EXPERIENCE

Amazon, Inc. (AWS Timestream team)
Software Development Engineer (SDE) Intern; Manager: Audrey Lawrence — June – Aug. 2023

- Designed and implemented distributed caches for user authentication in Timestream’s data ingestion routers, leveraging Amazon ElastiCache and Key Management Service (KMS). Reduced auth server calls by 100x and significantly decreased system latency, securing a full-time return offer.

PUBLICATIONS

1. Diedrich, L. *, Tzouanas, C. *, **Pan, B.** *, Sherman, M., Goessling, W., Shalek., A. (2026). Condition-informed identification of gene expression programs from single-cell data with scCoral. Manuscript in preparation for *Nature Methods.*
2. Leshchiner, E.*, **Pan, B.** *, Lehotzky, D. *, George, A, Kamburov, A., Hess, J., Dunford, A., Getz, G. (2025). Mutational enrichment in 3D protein interfaces uncovers cancer driver in RNA regulatory pathway. Manuscript in preparation for submission to *Nature Cancer*.
3. **Pan, B**., Kwon, Y., Bagnato-Colin, E., Kelley, D. (2023). Ethology and evolution of courtship vocalization in Xenopus. *Society for Integrative and Comparative Biology (SICB) Annual Meeting 2023*. (Abstract)

4. Xue, Y., Yang, F., Li, J., Zuo, X., **Pan, B.**, Li, M., Quinto, L., Mehta, J., Stiefel, L., Kimmey, C., Eshed, Y., Zussman, E., Simon, M., & Rafailovich, M. (2021). Synthesis of an effective flame-retardant hydrogel for skin protection using xanthan gum and resorcinol bis(diphenyl phosphate)-coated starch. *Biomacromolecules*, *22*(11), 4535–4543. https://doi.org/10.1021/acs.biomac.1c00804
5. **Pan, B.**, Wang, Y., Li, H., Yi, W., & Pan, Y. (2020). Preparation and electrosorption desalination performance of peanut shell-based activated carbon and MOS_2. *International Journal of Electrochemical Science*, *15*, 1861–1880. https://doi.org/10.20964/2019.12.74
6. Islam, D., Uddin, M. H., **Pan, B.**, & Joy, M. M. A. (2020). Flexible and high-energy dense yarn-shaped supercapacitor based on Ni-carbon nanotubes framework. *Chemical Physics Letters*, *760*, 138007. https://doi.org/10.1016/j.cplett.2020.138007
7. Xiao, X., Lin, Y., **Pan, B**., Fan, W., & Huang, Y. (2018). Photocatalytic degradation of methyl orange by $BiOI/Bi_4O_5I_2$ microspheres under visible light irradiation. *Inorganic Chemistry Communications*, *93*, 65–68. https://doi.org/10.1016/j.inoche.2018.05.009

MEETINGS AND PRESENTATIONS

1. Joint Meeting of the St. Jude Research Collaboratives, Memphis, TN, April 2026. (Oral)
2. Society for Integrative and Comparative Biology (SICB) Meeting, Austin, TX, January 2023. (Poster)
3. Shanghai Science Festival 2021, Shanghai, China, May 2021. (One of four Teen Innovators invited)
4. Sigma Xi 2020 Annual Meeting & Student Research Conference, Online, November 2020. (Oral)
5. American Chemical Society (ACS) Spring 2020 Meeting, Philadelphia, PA, March 2020. (Poster)
6. Materials Research Society (MRS) Fall 2019 Meeting, Boston, MA, December 2019. (Poster)

GRANTS AND AWARDS

1. Departmental Computer Science Scholarship Award, Columbia University (2024; one of two students awarded)
2. Rhodes Scholarship Finalist (2023)
3. Upsilon Pi Epsilon (UPE) International Computer Science Honor Society Inductee (2023)
4. Robert K. Kraft Global Fellowship (2022)
5. Ng Teng Fong Internship Award (2021)
6. International Science and Engineering Fair (ISEF) Finalist (2020)

LEADERSHIP/SERVICE

Harvard-MIT Health Sciences and Technology Student Association. *Social Chair* Sept. 2024 – Present

- Lead social programming for the HST and broader MIT biomedical community; organize monthly department dinners and annual signature events (e.g. ice skating and formal gala), engaging 300+ students.

Columbia College Student Council. *Class of 2024 Representative* Oct. 2022 – May 2024

- Elected by peers to represent the class. Spearheaded university-wide events including ‘After Hour at Luna Park’ and ‘Explore NYC Initiative for CC’ engaging 2000+ students. Actively contributed to constitutional reform and student wellbeing policy recommendations.

Columbia University Systems Biology Initiative. *Vice President*, *Podcast Host* Nov. 2021 – May 2024

- Produced podcast series Bio Bytes and BioWorks to communicate systems biology to public. Oversaw club event promotions and finances, and organized visits to biotech startup incubators.

OTHERS

Languages: Mandarin (Native), English (Bilingual), Cantonese (Conversational)

Computer Programming: Python, Java, C, Julia, R, LaTeX, Arduino IDE, Bash, GitHub

Interests: Basketball (Columbia Intramural D3 Second Place, Spring 2022), Cycling (Orlando -> Key West, 753 km, May 2022), Writing (Medium page)